

ESP32 with DHT22 Components Table

Component Name	Quantity	Value	Purpose
Capacitors (C2, C3, C4, C6, C7, C8, C9, C10, C11, C12, C13)	11	100nF	Decoupling and noise filtering.
Capacitor (C1)	1	22nF	Signal smoothing and filtering.
Capacitor (C5)	1	4.7uF	Power stabilization.
Resistor (R2)	1	560Ω	Current limiting for LED protection.
Resistor (R3)	1	22.1kΩ	Voltage division and logic control.
Resistor (R4)	1	47.5kΩ	Voltage divider/signal conditioning
Resistors (R5, R12)	2	1kΩ	Current limiting and component protection.
Resistors (R1, R8, R9)	3	5.1kΩ	USB-C configuration and current control.
Resistors (R6, R7, R10, R11, R16, R17)	6	27Ω	USB data line signal integrity.
Resistors (R13, R14, R15, R18, R19, R20)	6	10kΩ	Pull-up resistors for logic stability.
D1	1	19-219/GHC-YR2T1B5Y/3T	Power/status indication LED.
D2	1	NCD0603R6	Activity/status indication LED.
D3	1	B5819W-TP	Reverse polarity protection.
D4, D5, D6	3	LESD5D5.0CT1G-ES	Electrostatic discharge (ESD) protection.
J1	1	X1311WV-03J-C40D24	Power input selection
J2	1	GT-USB-7010ASV	USB communication and power interface.
J3	1	DZ127R-11-02-25	External connection interface.

J4, J5	2	X6511WV-17H-C60D30	External hardware connection interfaces.
Q1, Q2	2	2N7002LT1G	Auto-reset and boot mode switching.
Q3, Q4	2	PBSS4120T,215	Signal switching/control
Switches (SW1, SW2)	2	TSA063G60-250	Manual reset and boot control.
ESP32 (U1)	1	ESP32-WROOM-32E-N8	Main microcontroller and processing unit.
U2	1	NCP1117LPST33T3G	5V to 3.3V voltage regulation.
U3	1	CP2102N-A02-GQFN28R	USB to UART bridge.
DHT22 (U4)	1	DHT22	Temperature and humidity sensing.